

An Analysis of Gaussian process

Gaussian process

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Gaussian process -- Part 1

GPML: A comprehensive Matlab toolbox for GP regression and classification
STK: a Small (Matlab/Octave) Toolbox for Kriging and GP modeling
Kriging module in UQLab framework (Matlab)
CODES Toolbox: implementations of Kriging, variational kriging and multi-fidelity models (Matlab)
Matlab/Octave function for stationary Gaussian fields
Yelp MOE – A black box optimization engine using Gaussian process learning
ooDACE Archived 2020-08-09 at the Wayback Machine – A flexible object-oriented Kriging Matlab toolbox.
GPstuff – Gaussian process toolbox for Matlab and Octave
GPy – A Gaussian processes framework in Python
GSTools - A geostatistical toolbox, including Gaussian process regression, written in Python
Interactive Gaussian process regression demo
Basic Gaussian process library written in C++
scikit-learn – A machine learning library for Python which includes Gaussian process regression and classification
SAMBO Optimization library for Python supports sequential optimization driven by Gaussian process regressor from scikit-learn. [1] - The Kriging toolKit (KriKit) is developed at the Institute of Bio- and Geosciences 1 (IBG-1) of Forschungszentrum Jülich (FZJ)

Gaussian process -- Part 2

Here $d = |x - x'|$, which is a result of the stationary process property, that is, for any stationary process the covariance function will only depend on d . The parameter ℓ is the characteristic length-scale of the process (practically, "how close" two points x and x' have to be to influence each other significantly), δ is the Kronecker delta and σ the standard deviation of the noise fluctuations. Moreover, K_ν is the modified Bessel function of order ν and $\Gamma(\nu)$ is the gamma function evaluated at ν . Importantly, a complicated covariance function can be defined as a linear combination of other simpler covariance functions in order to incorporate different insights about the data-set at hand. The inferential results are dependent on the values of the hyperparameters θ (e.g. ℓ and σ) defining the model's behaviour. A popular choice for θ is to provide maximum a posteriori (MAP) estimates of it with some chosen prior. If the prior is very near uniform, this is the same as

maximizing the marginal likelihood of the process; the marginalization being done over the observed process values y . This approach is also known as maximum likelihood II, evidence maximization, or empirical Bayes.

Gaussian process -- Part 3

A key fact of Gaussian processes is that they can be completely defined by their second-order statistics. Thus, if a Gaussian process is assumed to have mean zero, defining the covariance function completely defines the process' behaviour. Importantly the non-negative definiteness of this function enables its spectral decomposition using the Karhunen–Loève expansion. Basic aspects that can be defined through the covariance function are the process' stationarity, isotropy, smoothness and periodicity. Stationarity refers to the process' behaviour regarding the separation of any two points x and x' . If the process is stationary, the covariance function depends only on $x - x'$. For example, the Ornstein–Uhlenbeck process is stationary. If the process depends only on $|x - x'|$, the Euclidean distance (not the direction) between x and x' , then the process is considered isotropic. A process that is concurrently stationary and isotropic is considered to be homogeneous; in practice these properties reflect the differences (or rather the lack of them) in the behaviour of the process given the location of the observer. Ultimately Gaussian processes translate as taking priors on functions and the smoothness of these priors can be induced by the covariance function. If we expect that for "near-by" input points x and x' their corresponding output points y and y' to be "near-by" also, then the assumption of continuity is present. If we wish to allow for significant displacement then we might choose a rougher covariance function. Extreme examples of the behaviour is the Ornstein–Uhlenbeck covariance function and the squared exponential where the former is never differentiable and the latter infinitely differentiable. Periodicity refers to inducing periodic patterns within the behaviour of the process. Formally, this is achieved by mapping the input x to a two dimensional vector $u(x) = (\cos(x), \sin(x))$.

Gaussian process -- Part 4

and maximizing this marginal likelihood towards θ provides the complete specification of the Gaussian process f . One can briefly note at this point that the first term corresponds to a penalty term for a model's failure to fit observed values and the second term to

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a penalty term that increases proportionally to a model's complexity. Having specified θ , making predictions about unobserved values $f(x^*)$ at coordinates x^* is then only a matter of drawing samples from the predictive distribution $p(y^* | x^*, f(x), x) = N(y^* | A, B)$ where the posterior mean estimate A is defined as

Gaussian process -- Part 5

In these two cases the function σ is increasing on $[0, \infty)$, but generally it is not. Moreover, the condition

Gaussian process -- Part 6

$f(x) = G X g \sim G P(G X \mu g, G X K g G X' T)$.

$$f(x) = \{G\}_{X} g \sim \{G\}_{X} \mu_{g}, \{G\}_{X} K_{g} \{G\}_{X'} T$$

Gaussian process -- Part 7

$\log p(f(x') | \theta, x) = -\frac{1}{2} (f^T(x) K^{-1}(\theta, x, x') f(x') + \log \det(K(\theta, x, x')) + n \log 2\pi)$

$$\log p(f(x') | \theta, x) = -\frac{1}{2} (f^T(x) K^{-1}(\theta, x, x') f(x') + \log \det(K(\theta, x, x')) + n \log 2\pi)$$